

20260921

Re-cap on the brms model results: super large credible interval for PSE estimates

→ Are these PSE estimates accurate?

[Checkpoint 1] brms model setup

References for analysis:

Separate model for exposure and test (from TJ2025)

VOT scaling, backward difference coding (from SI)

Model prior and setup (OSF code): (TJ-OSF)fit_test_upto_block4.R

```
fit_test_upto_block4 <-  
  brm(  
    bf(Response.Voiceless ~ 1,  
        mu1 ~ 0 + offset(0),  
        mu2 ~ 1 + VOT_gs * Condition.Exposure * Block +  
          (1 + VOT_gs) * Block | ParticipantID) + (1 + VOT_gs * Condition.Exposure * Block | Item.MinimalPair),  
    theta1 ~ 1),  
    data = d.IA_predicted_logodds,  
    prior = my_priors,  
    cores = 4,  
    chains = 4,  
    init = 0,  
    iter = 4000,  
    warmup = 2000,  
    save_pars = save_pars(all = TRUE),  
    family = mixture(bernoulli("logit"), bernoulli("logit"), order = F),  
    control = list(adapt_delta = 0.99),  
    file = "/home/mtan12/AEDLVOT/models/test-standard-SD15-upto-block4.rds",  
    file_refit = "on_change"  
  )
```

Many other self-compiled functions were not available in the OSF repository

[Checkpoint 2] Method to extract PSE estimates from the brms model

(old)

Exposure and test blocks

- (1) Establish a prediction grid: 3 exposure condition x 3 exposure block x 100 point across scaled VOT range
- (2) Use parameters from each of the 8000 posterior draws (4 chains x 2000 post-warmup iterations=8000) from the brms model (fit_exposure) to obtain predicted response probabilities across the entire grid, resulting in 3 x 3 x 100 x 8000 predictions
- (3) For each posterior draw, identify the VOT point with the predicted probability closest to p=0.5 as the PSE per exposure condition and exposure block, resulting in 3 x 3 x 8000 = 72000 PSE estimates
- (4) Derive the group-level median and 95% credible interval of the PSE estimates per exposure condition and exposure block, and unscaled the derived PSE back to ms

(new)

Exposure and test blocks

- (1) For each posterior draw per exposure condition and exposure block, use scaled VOT=0 and scale VOT=1 to derive intercept and slope on the log-odds scale (which is linear)
- (2) Derive PSE estimates by -intercept/slope (as mentioned the caption in TJ2025 Figure S7) per exposure condition and exposure block, and unscaled the derived PSE back to ms

Results: similar PSE estimates, similar large credible interval
(cf. 20260821 SpanishVOT-brms-models-and-PSE-predictions.pdf)

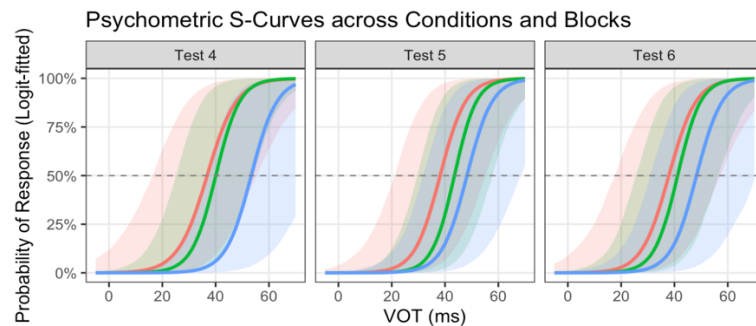
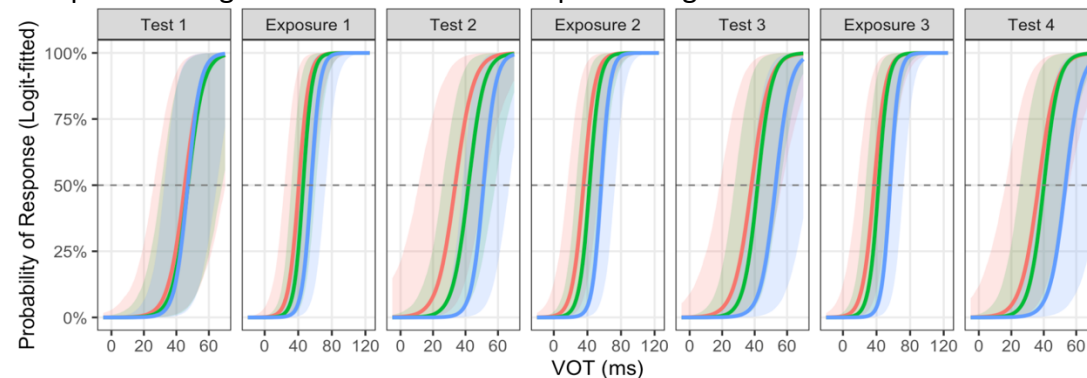
Old

participant.condition <fctr>	exp_stagelD <fctr>	group_PSE_median <dbl>	group_PSE_lower <dbl>	group_PSE_upper <dbl>	ci_width <dbl>
expo-20ms	Exposure 1	40.05051	23.93939	57.62626	33.68687
expo-20ms	Exposure 2	35.65657	16.61616	53.26894	36.65278
expo-20ms	Exposure 3	37.12121	16.61616	56.16162	39.54545
expo-10ms	Exposure 1	45.90909	31.26263	60.55556	29.29293
expo-10ms	Exposure 2	42.97980	26.86869	59.09091	32.22222
expo-10ms	Exposure 3	42.97980	26.86869	57.62626	30.75758
expo+20ms	Exposure 1	54.69697	40.05051	72.27273	32.22222
expo+20ms	Exposure 2	56.16162	42.94318	72.27273	29.32955
expo+20ms	Exposure 3	56.16162	41.51515	72.27273	30.75758

New

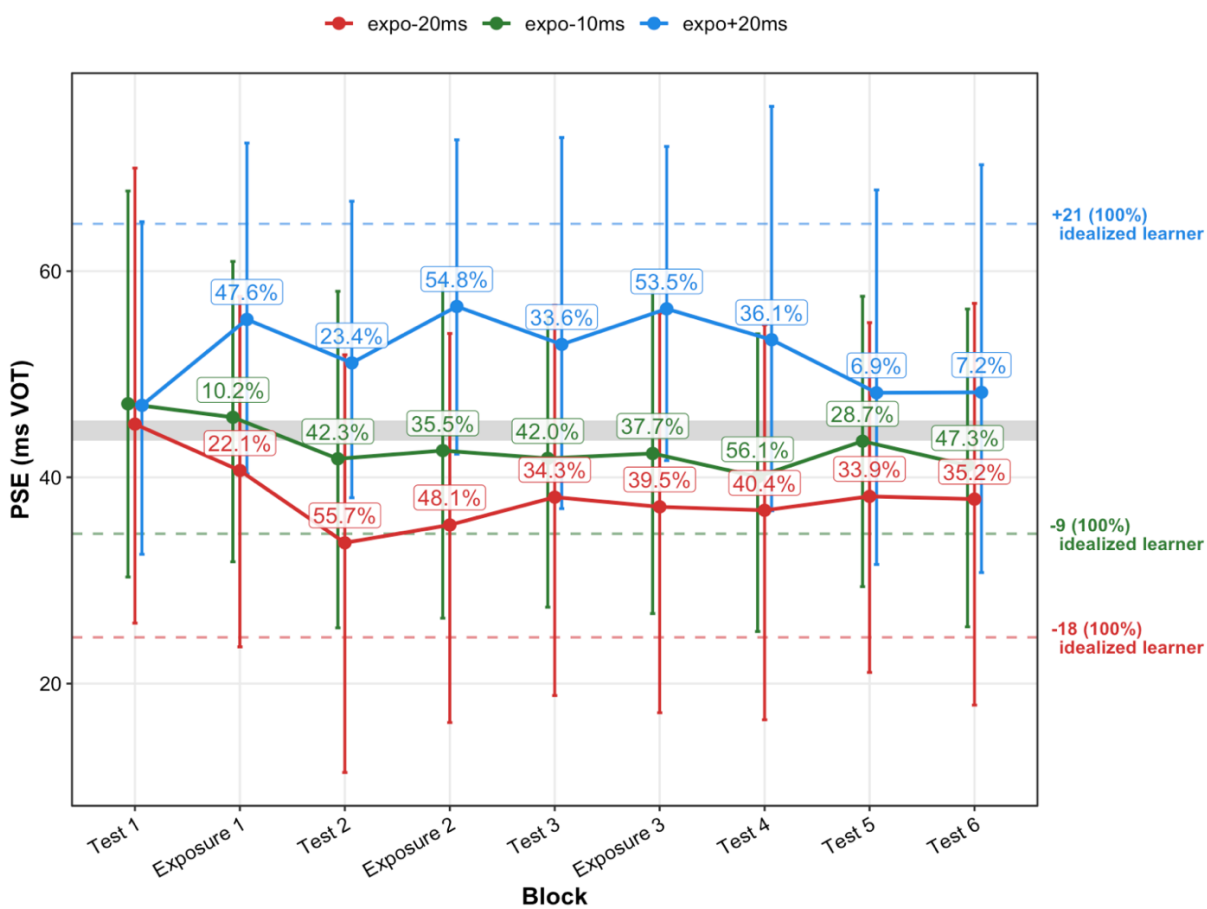
participant.condition <fctr>	exp_stagelD <fctr>	group_PSE_median <dbl>	group_PSE_lower <dbl>	group_PSE_upper <dbl>	ci_width <dbl>
expo-20ms	Exposure 1	40.66134	23.56533	57.53138	33.96605
expo-20ms	Exposure 2	35.38677	16.22550	53.95039	37.72489
expo-20ms	Exposure 3	37.14060	17.17295	55.94022	38.76727
expo-10ms	Exposure 1	45.82643	31.80907	60.93747	29.12841
expo-10ms	Exposure 2	42.59694	26.34392	58.98917	32.64525
expo-10ms	Exposure 3	42.31745	26.78872	58.15265	31.36393
expo+20ms	Exposure 1	55.31676	40.34412	72.42042	32.07630
expo+20ms	Exposure 2	56.57160	42.25197	72.73254	30.48056
expo+20ms	Exposure 3	56.34151	41.62618	72.08489	30.45870

Group-level categorization curves and PSE plots using new PSE estimates



Exposure Condition expo-20ms expo-10ms expo+20ms

Perceptual Boundary Shift across Experimental Blocks



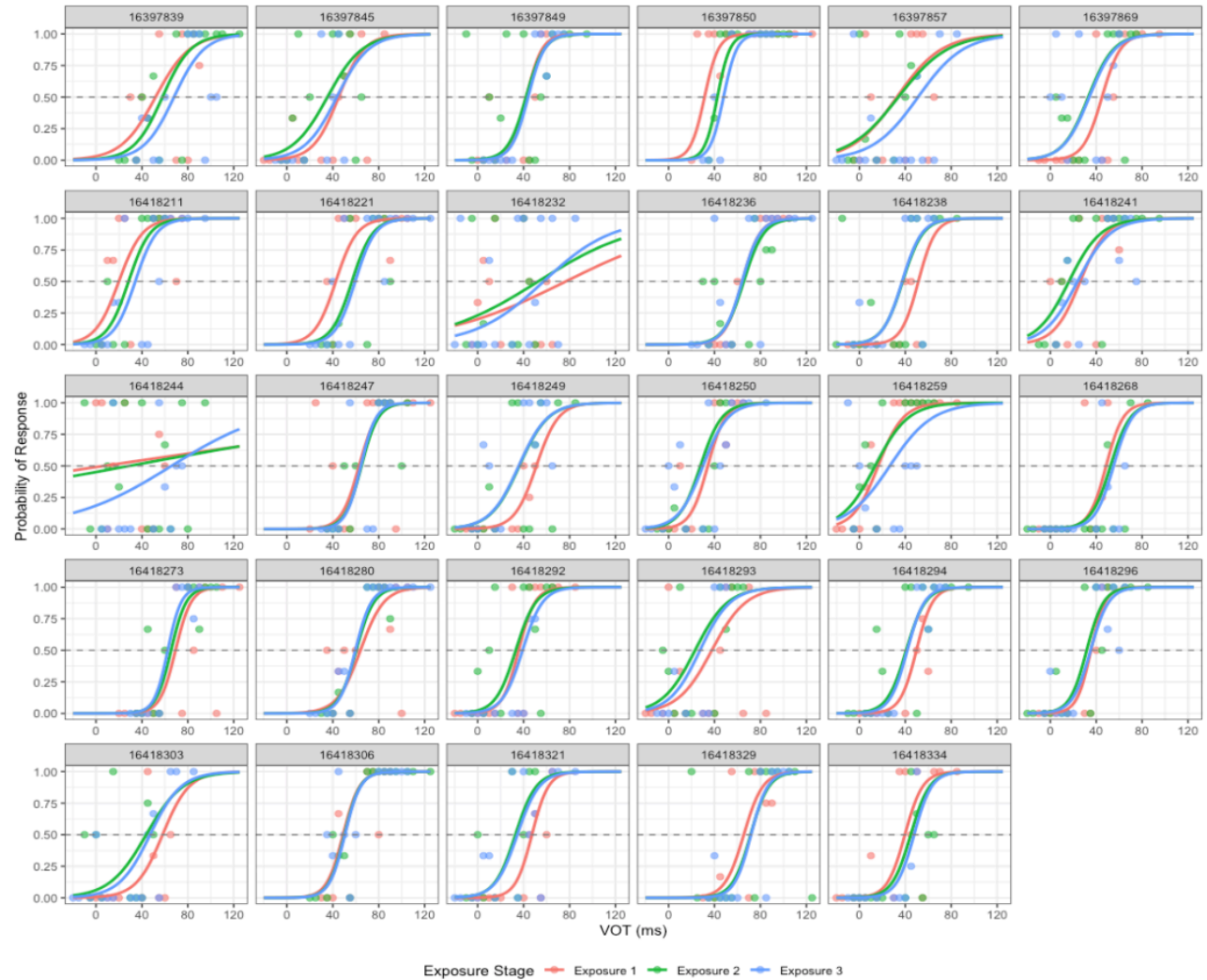
Use new PSE estimates to identify good and bad subjects:

Good (Crl < 25): N=0

Bad Crl (Crl > 50): N=29

Bad slope (VOT slope crossing 0): N=5

Bad subjects

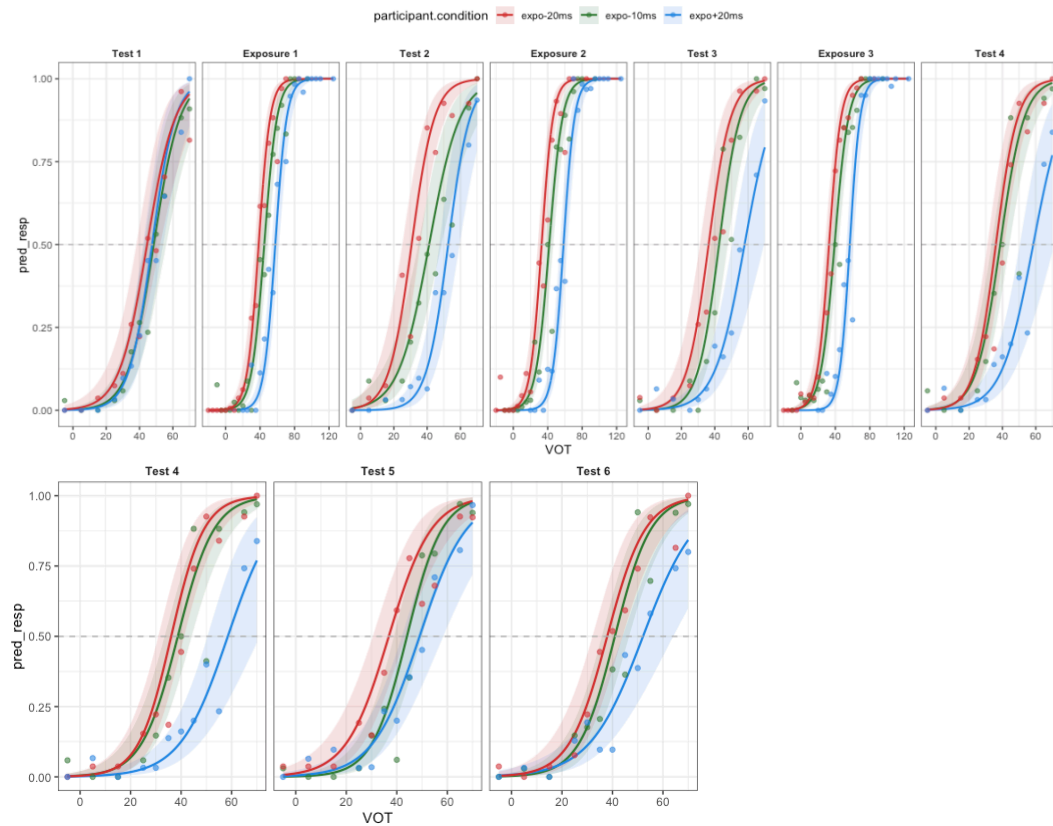


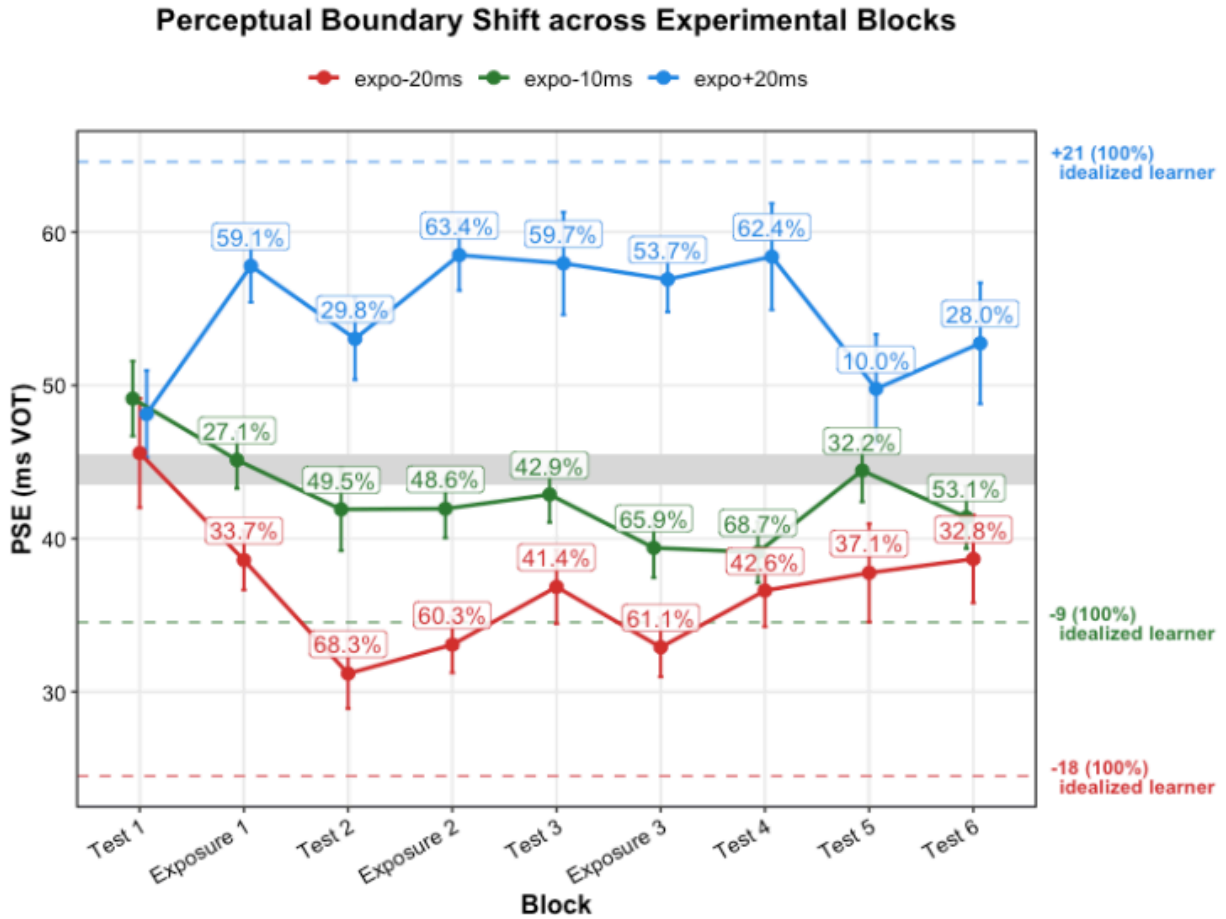
**individual-level categorization curve fixed – the slopes of exposure block were not properly included in the earlier analysis, but now the block effect can be observed

[Checkpoint 3] Exploratory analysis – LLM modeling excluding 29 bad subjects (i.e., highly uncertain PSE estimates and unstable VOT slopes in the Bayesian model)

- Compare with the full LLM conducted at the very beginning to see if the group-level PSE median and 95% CrI drastically changed (cf. 20260805 SpanishVOT-preliminary-analysis.pdf)
- Although not considered the lapse rate, TJ2025 replicated all the brms results using LLM analysis

N=92





→ while the proportion of adaptation appears to be overall greater excluding bad subjects, the qualitative pattern looks similar to the LLM analysis with the full dataset